Hyuk Che (Luke) Kwon

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Education

University of Pennsylvania | Philadelphia, PA — Aug. 2025 - May 2027
Master of Science in Engineering, Computer Graphics and Game Technology — GPA: 3.95

Sogang University | Seoul, South Korea — Mar. 2018 - Aug. 2024
Bachelor of Science in Engineering, Computer Science and Engineering — Major GPA: 3.92
Bachelor of Arts, Global Korean Studies — Magna Cum Laude

Programming Skills

Languages: C, C++, GLSL, WGSL, Python, TypeScript, F#

Tools/Frameworks: CUDA, OptiX, OpenGL, WebGPU, NVIDIA Nsight, RenderDoc, GLM, Maya, Blender, Houdini, Qt

Experience

Software Engineering Intern | *VS Graphics Team, LG Electronics* — Jun. 2026 - Aug. 2026

- Optimized CARLA, an Unreal Engine driving simulator for training autonomous vehicles in virtual environments
- Multithreaded JPEG decoding and removed redundant mipmaps for texture streaming, raising FPS from 11 to 20
- Diagnosed a GPU stall via Unreal Insights and implemented asynchronous GPU readback, raising FPS from 20 to 30
- Developed a Python script for dynamic client-side FPS pacing and diagnosed a memory leak in no-rendering mode

Teaching Assistant | *University of Pennsylvania* — Jan. 2026 - Present

- Assisted in teaching Introduction to Computer Graphics (CIS 5600) and Computer Animation (CIS 5620)
- Led review sessions, graded assignments, and held weekly office hours to help students with course material and projects

Projects

OptiX/CUDA Path Tracer | C++, CUDA, NVIDIA OptiX, Nsight Systems — Sep. - Oct. 2026

- Developed a hardware-accelerated path tracer supporting glTF meshes, GGX materials, texture/normal mapping, and Fresnel reflection/refraction
- Profiled GPU execution strategies with Nsight Systems, evaluating stream compaction and consolidating tracing and shading into OptiX raygen for up to 3× higher throughput
- Implemented HDR environment lighting with MIS, stochastic antialiasing, and thin-lens depth of field

WebGPU Forward+ / Clustered Deferred Renderer | TypeScript, WGSL, WebGPU — Oct. 2026

- Built Forward+ and clustered deferred pipelines with light clustering using compute shaders, achieving up to an 11.1× GPU speedup over naive forward rendering at 3,000 lights
- Implemented logarithmic depth slicing to allocate finer clusters near the camera and reduce light-list overflow
- Benchmarked GPU passes to analyze light counts, cluster resolutions, workgroup sizes, and light-list capacity tradeoffs

CUDA Boids Simulation | C++, CUDA, OpenGL, Nsight Systems — Sep. 2026

- Developed a CUDA flocking simulation with brute-force, scattered-grid, and coherent-grid neighbor search algorithms
- Replaced all-pairs search with uniform-grid partitioning and contiguous boid storage for efficient neighbor queries
- Profiled block sizes and grid granularities, raising kernel occupancy from 33% to 83% and simulation throughput by 17%

Mini-Minecraft | C++, OpenGL, GLSL, RenderDoc — Dec. 2025

- Implemented player physics by incorporating ray marching to support collision detection on spherical environments
- Sampled 3D Perlin noise for cave systems and built a post-processing pipeline for underwater effects
- Created a procedural skybox using 3D Worley noise to generate stars and 3D Perlin noise-based FBM for nebulae